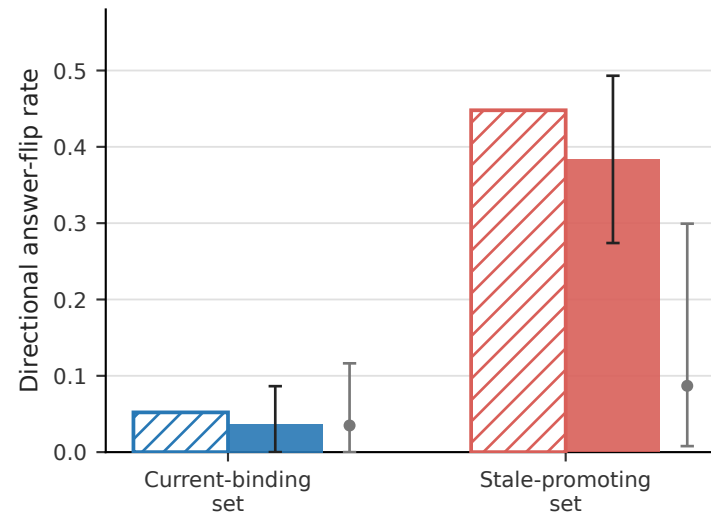
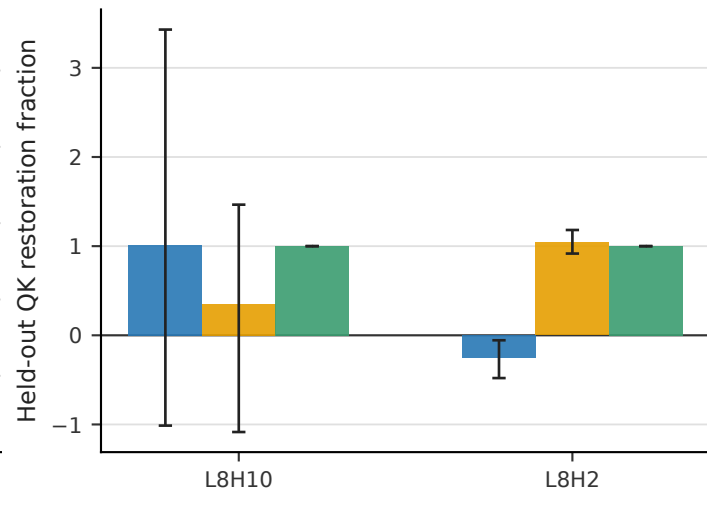
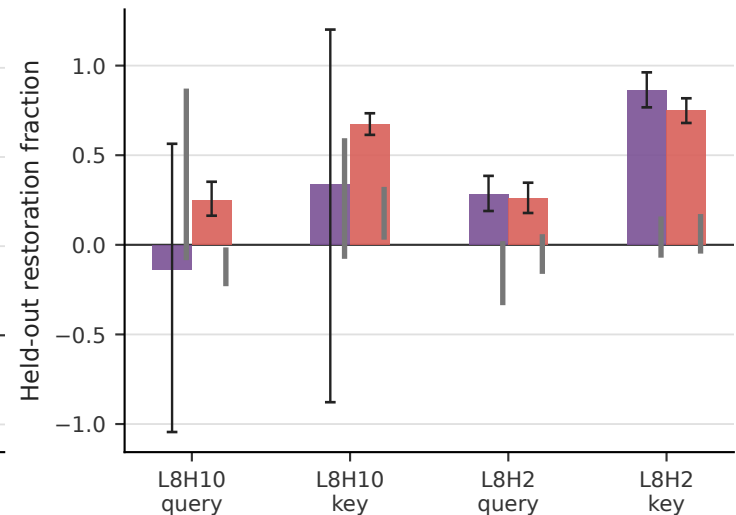


A Held-out ablation de-bias**B Query-key decomposition****C Discovery top-12 vs matched random**

Phase 1 same-pool
Phase 1b held-out

Matched random 95% range

Query

Key

Both

QK margin

Final gap